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ROLL NO:SA263
DIV:SA2
PROJECT TOPIC:“Netflix Content Analysis.
DATASET NAME:Netflix Dataset
SOURCE:PROVIDED DATASET FILE (.xls) FROM KAGGLE
NUMBER OF ROWS: MORE THAN 1000
NUMBER OF COLUMNS:10
TYPES OF DATA:

1. Categorical Data (Qualitative Data)

This type of data represents labels or categories, not numbers.

2. Numerical Data (Quantitative Data)

This data consists of numbers and can be used for mathematical calculations.

3. Discrete Data

A type of numerical data that takes specific fixed values (integers).

4. Text Data (Unstructured Data)

This is free-form text, not organized into numbers or categories directly.

5. Date/Time Data

This represents time-related information.

DESCRIPTION:
This project focuses on analyzing a real-world Netflix dataset to understand trends and patte

The dataset contains both categorical and numerical data, which allows for meaningful insight

The main goal of this project is to identify patterns in Netflix content and provide useful i
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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# for model
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score
```

```
df = pd.read_csv("netflix1-1.csv")

df.head()
```

	show_id	type	title	director	country	date_added	release_year	rating	duration
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	United States	9/25/2021	2020	PG-13	90 min
1	s3	TV Show	Ganglands	Julien Leclercq	France	9/24/2021	2021	TV-MA	1 Season
2	s6	TV Show	Midnight Mass	Mike Flanagan	United States	9/24/2021	2021	TV-MA	1 Season
3	s14	Movie	Confessions of an Invisible Girl	Bruno Garotti	Brazil	9/22/2021	2021	TV-PG	91 min
4	s8	Movie	Sankofa	Haile Gerima	United States	9/24/2021	1993	TV-MA	125 min

Next steps: Generate code with df New interactive sheet

```
print("Shape:", df.shape)
print("\nData Types:\n", df.dtypes)
```

```
Shape: (8790, 10)

Data Types:
 show_id         object
type            object
title           object
director        object
country         object
date_added      object
release_year     int64
rating          object
duration        object
listed_in       object
dtype: object
```

```
df.isnull().sum()
```

	0
show_id	0
type	0
title	0
director	0
country	0
date_added	0
release_year	0
rating	0
duration	0
listed_in	0

dtype: int64

```
# Fill categorical columns with 'Unknown'
df.fillna("Unknown", inplace=True)
```

```
df.drop_duplicates(inplace=True)
```

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### Data Cleaning Summary

#The dataset had missing values in columns like director, cast, and country.
#Instead of dropping rows (which would lose data), I filled them with "Unknown".
#Duplicate rows were removed to ensure clean analysis.
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```
# Using release_year as numeric column
col = df['release_year']

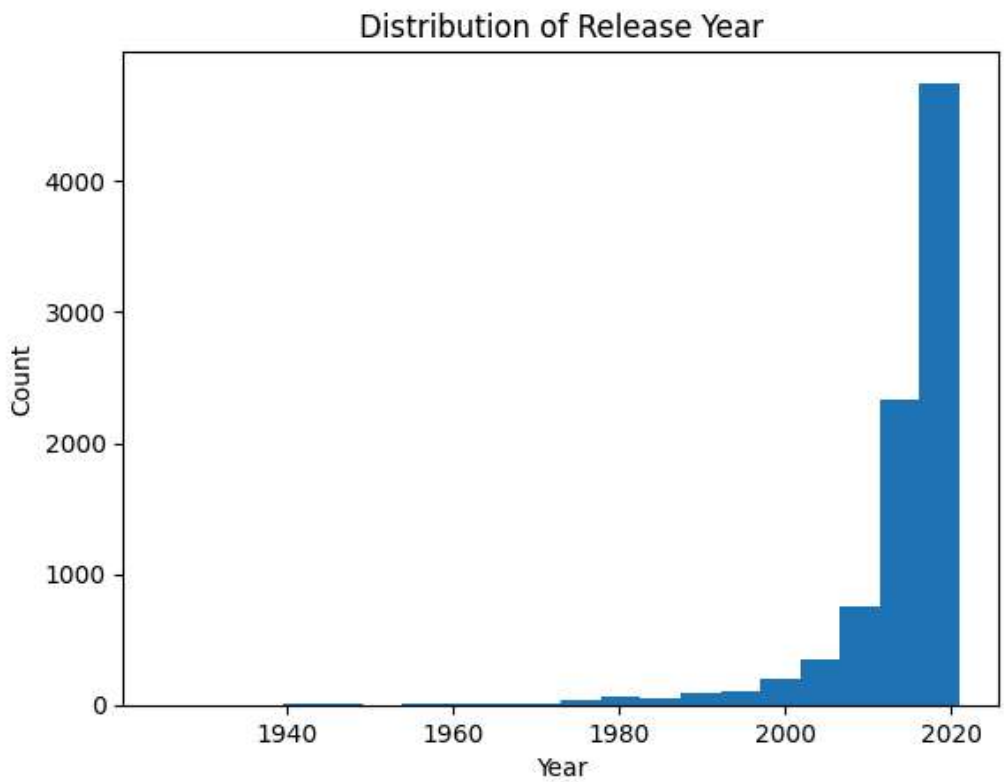
mean = col.mean()
median = col.median()
mode = col.mode()[0]
std = col.std()
var = col.var()
range_val = col.max() - col.min()
mid_range = (col.max() + col.min()) / 2

print("Mean:", mean)
print("Median:", median)
print("Mode:", mode)
print("Standard Deviation:", std)
print("Variance:", var)
print("Range:", range_val)
print("Mid-range:", mid_range)
```

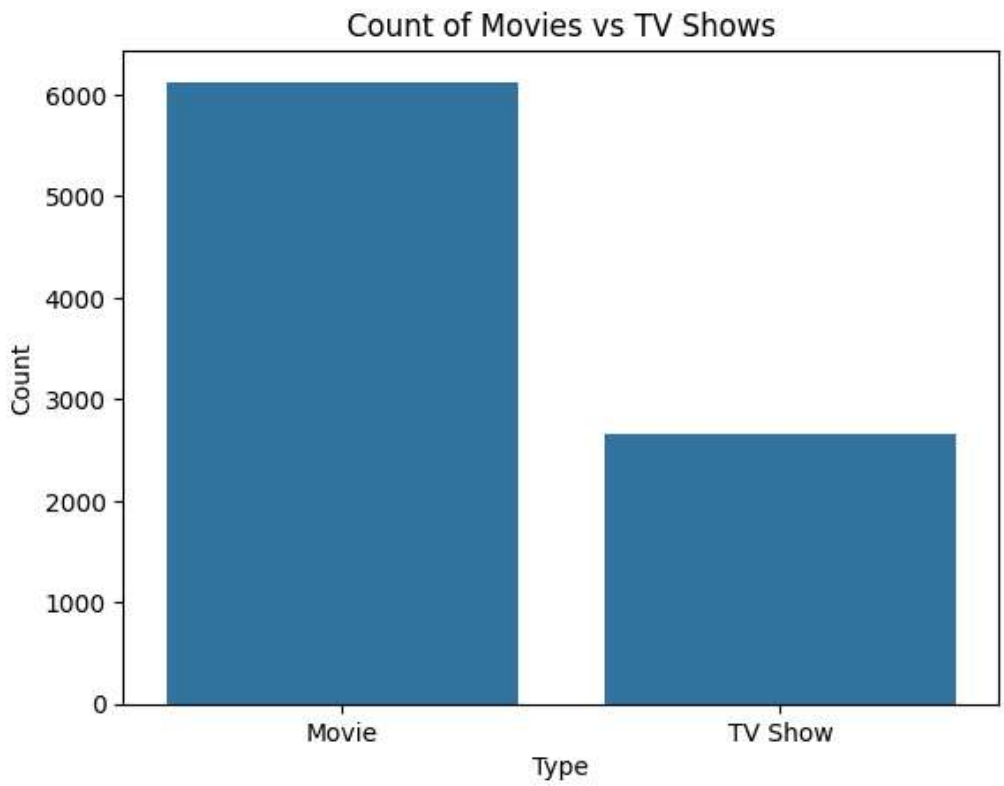
```
Mean: 2014.183162684869
Median: 2017.0
Mode: 2018
Standard Deviation: 8.82546602834756
Variance: 77.88885061751685
```

```
Range: 96
Mid-range: 1973.0
```

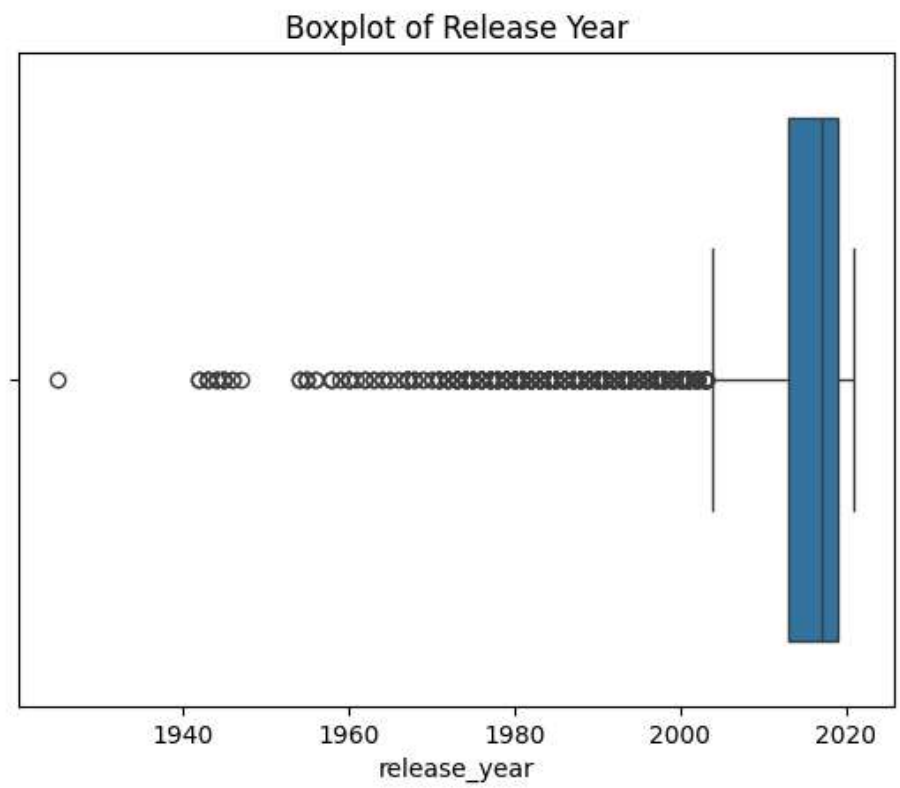
```
plt.hist(df['release_year'], bins=20)
plt.title("Distribution of Release Year")
plt.xlabel("Year")
plt.ylabel("Count")
plt.show()
```




```
sns.countplot(x='type', data=df)
plt.title("Count of Movies vs TV Shows")
plt.xlabel("Type")
plt.ylabel("Count")
plt.show()
#The chart shows that movies are more common than TV shows on Netflix.
```

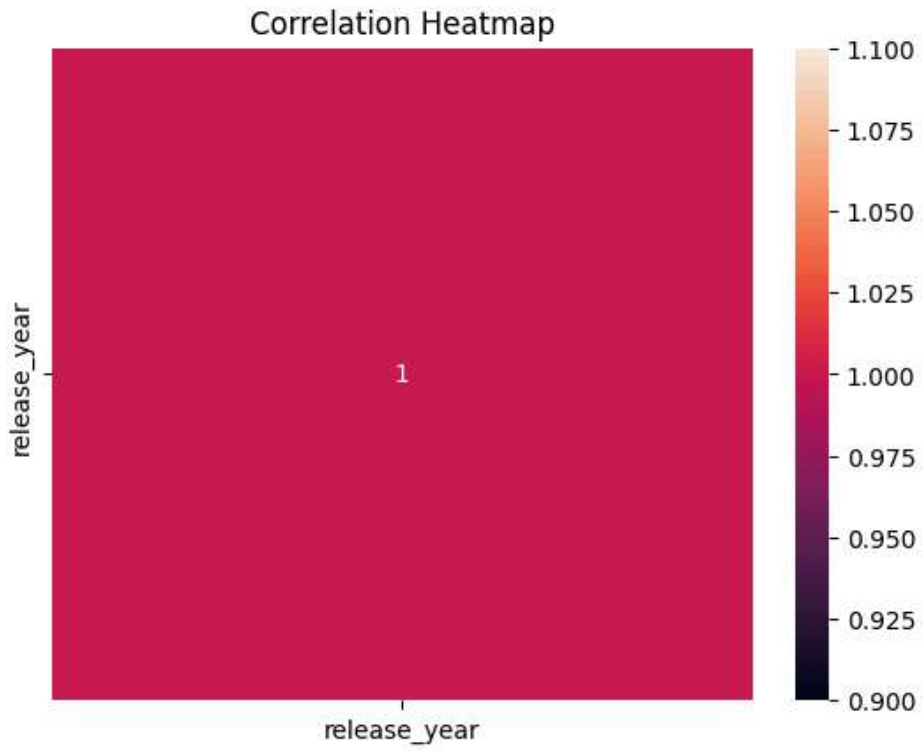



```
sns.boxplot(x=df['release_year'])
plt.title("Boxplot of Release Year")
plt.show()
# The boxplot shows some older content as outliers, but most data is
# concentrated in recent years.
```



```
# Select numeric columns
numeric_df = df.select_dtypes(include=['int64', 'float64'])

sns.heatmap(numeric_df.corr(), annot=True)
plt.title("Correlation Heatmap")
plt.show()
```




```
le = LabelEncoder()

df['type_encoded'] = le.fit_transform(df['type'])
df['rating_encoded'] = le.fit_transform(df['rating'])
```

```
X = df[['release_year', 'rating_encoded']]
y = df['type_encoded']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

```
model = DecisionTreeClassifier()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
#The model tries to predict whether a title is a Movie or TV Show based on release year and
#The accuracy shows how well the model performs. A higher accuracy means better prediction c
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```
Accuracy: 0.7201365187713311
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### Insights

#1. Most Netflix content is released after 2000 (Histogram).
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#2. Movies are significantly more than TV shows (Bar Chart).
#3. Older shows appear as outliers (Boxplot).

### Recommendations

#1. Netflix should continue focusing on recent content as it dominates the platform.
#2. Increasing TV show content may help balance the catalog and attract binge-watch users.
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# Netflix Data Analysis Project

## Description
#This project analyzes Netflix dataset to understand trends in content type, release years,

## Files
#- netflix1-1.csv
#- netflix_analysis.ipynb
```